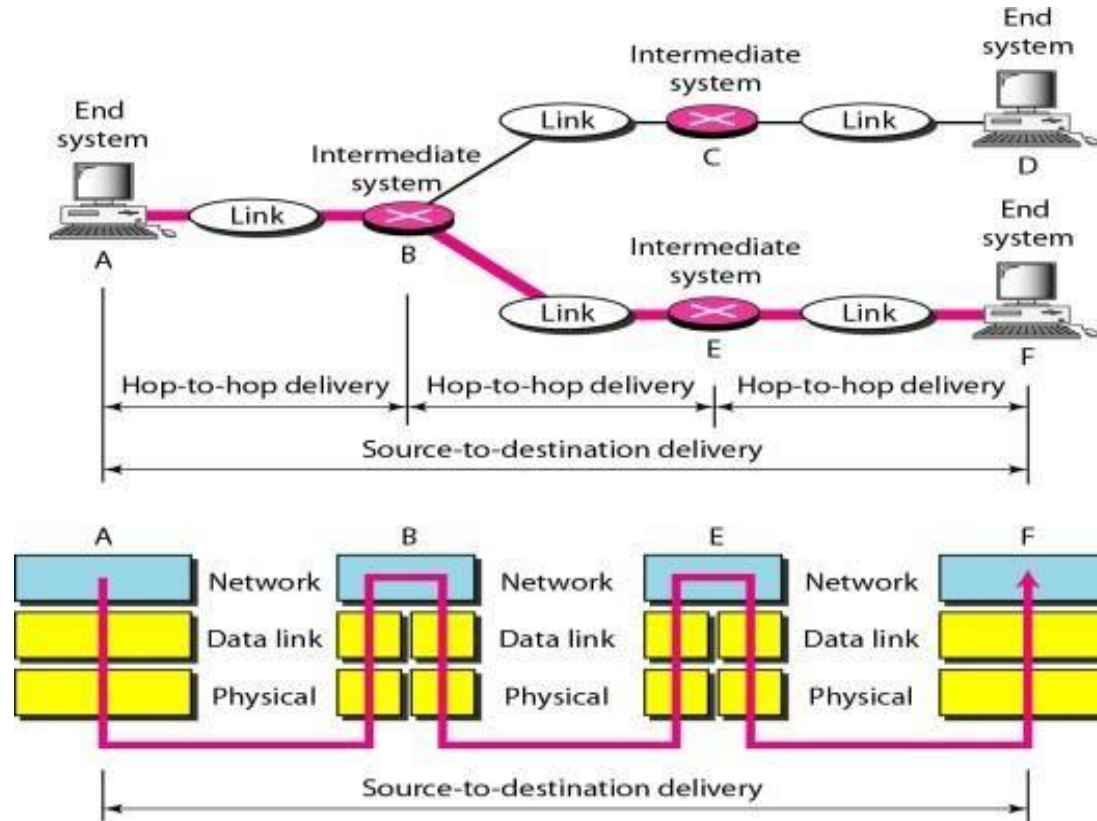


Module :3 The NetworkLayer

Source-to-destination delivery



Forwarding and routing functions of the network layer

- Forwarding involves the transfer of a packet from an incoming link to an outgoing link within a *single* router.
- Routing means, the network layer must determine the route or path taken by packets as they flow from a sender to a receiver. The algorithms that calculate these paths are referred to as **routing algorithms**.
- The primary role of the routers is to forward datagrams from input links to output links. Every router has a **forwarding table**.
 - ✓ A router forwards a packet by examining the value of a field in the arriving packet's header, and then using this header value to index into the router's forwarding table.

✓ The **value stored in the forwarding** table entry for that header indicates the **router's outgoing link** interface to which that packet is to be forwarded.

✓ **Depending on the network-layer protocol**, the header value could be the destination address of the packet or an indication of the connection to which the packet belongs.

Some possible services that the network layer

- ***Guaranteed delivery with bounded delay.*** This service not only guarantees delivery of the packet, but delivery within a specified host-to-host delay bound
- ***In-order packet delivery.*** This service guarantees that packets arrive at the destination in the order that they were sent.

- ***Guaranteed minimal bandwidth.*** This network-layer service emulates the behavior of a transmission link of a specified bit rate.
- ***Guaranteed maximum jitter.*** This service guarantees that the amount of time between the transmission of two successive packets at the sender is equal to the amount of time between their receipt at the destination.
- ***Security services.*** Using a secret session key known only by a source and destination host, the network layer in the source host could encrypt the payloads of all datagrams being sent to the destination host.
- **best-effort service-** With best-effort service, timing between packets is not guaranteed to be preserved, packets are not guaranteed to be received in the order in which they were sent, nor is the eventual delivery of transmitted packets guaranteed.

Virtual Circuit and Datagram Networks

Like Transport layer , Network-layer offer connection oriented and connectionless services

- Computer networks that provide only a connection service at the network layer are called **virtual-circuit (VC) networks**; computer networks that provide only a connectionless service at the network layer are called **datagram networks**.

How a VC service can be implemented in a computer network ?

A VC consists of

- (1) a path (that is, a series of links and routers) between the source and destination hosts,
- (2) VC numbers, one number for each link along the path, and
- (3) entries in the forwarding table in each router along the path.

- A packet belonging to a virtual circuit will carry a VC number in its header.

Suppose that Host A requests that the network establish a VC between itself and Host B.

Also the network chooses the path A-R1-R2-B and assigns VC numbers 12, 22, and 32 to the three links in this path for this virtual circuit.

In this case, when a packet in this VC leaves Host A, the value in the VC number field in the packet header is 12; when it leaves R1, the value is 22; and when it leaves R2, the value is 32.

Whenever a new VC is established across a router, an entry is added to the forwarding table. Similarly, whenever a VC terminates, the appropriate entries in each table along its path are removed.

- In a VC network, the network's routers must maintain **connection state information** for the ongoing connections.
- Specifically, each time a new connection is established across a router, a new connection entry must be added to the router's forwarding table; and each time a connection is released, an entry must be removed from the table.

There are three identifiable phases in a virtual circuit

1. VC setup.

2. Data transfer.

3. VC teardown.

VC setup: The network layer determines the path between sender and receiver, that is, the series of links and routers through which all packets of the VC will travel.

The network layer also determines the VC number for each link along the path. Finally, the network layer adds an entry in the forwarding table in each router along the path.

Data transfer: once the VC has been established, packets can begin to flow along the VC.

VC teardown: This is initiated when the sender (or receiver) informs the network layer of its desire to terminate the VC.

- The messages that the end systems send into the network to initiate or terminate a VC, and the messages passed between the routers to set up the VC (that is, to modify connection state in router tables) are known as **signaling messages**, and the protocols used to exchange these messages are often referred to as **signaling protocols**.

Datagram Networks

Computer networks that provide connection-oriented services are called Virtual Circuits while those providing connection-less services are called **Datagram networks**.

In a **datagram network**, each packet is stamped with destination address.

As a packet is transmitted from source to destination, it passes through a series of routers. Each of these routers uses the packet's destination address to forward the packet.

When a packet arrives at the router, the router uses the packet's destination address to look up the appropriate output link interface in the forwarding table.

The router then intentionally forwards the packet to that output link interface.

Difference between Virtual Circuits and Datagram Networks

The following are the important differences between Virtual Circuits and Datagram Networks –

Key	Virtual Circuits	Datagram Networks
Definition	Virtual Circuit is a connection-oriented service in which there is an implementation of resources like buffers, CPU, bandwidth, etc., used by virtual circuit for a data transfer session.	Datagram networks are a type of connectionless service where no such resources are required for data transmission.
Path	In Virtual circuits, as all the resources and bandwidth get reserved before the transmission, the path which is utilized or followed by first data packet would get fixed and all other data packets will use the same path and consume same resources.	In a Datagram network, the path is not fixed as data packets are free to decide the path on any intermediate router on the go by dynamically changing routing tables on routers.
Header	As there is same path followed by all the data packets, a common and same header is	Different headers with information of other data packet is being used in

	being used by all the packets.	Datagram network.
Complexity	Virtual Circuit is less complex as compared to that of Datagram network.	Datagram network are more complex as compared to Virtual circuit.
Reliability	Due to fixed path and assurance of fixed resources, Virtual Circuits are more reliable for data transmission as compared to Datagram network.	Datagram networks, due to their dynamic resource allocation and dynamic path, are more errorprone and less reliable than Virtual circuits.
Example and Cost	Virtual circuits are costlier in installation and maintenance. They are widely used by ATM (Asynchronous Transfer Mode) Network, which is used for the Telephone calls.	Datagram networks are cheaper as compared to the Virtual Circuits. They are mainly used by IP network, which is used for Data services like Internet.

Network-Layer Services

- **Packetizing**

The first duty of the network layer is definitely packetizing: encapsulating the payload(data received from upper layer) in a network-layer packet at the source and decapsulating the payload from the network-layer packet at the destination.

- **Routing**

Another duty of the network layer, which is as important as the first, is routing. The network layer is responsible for routing the packet from its source to the destination.

- **Forwarding**

If routing is applying strategies and running some routing protocols to create the decision-making tables for each router, forwarding can be defined as the action applied by each router when a packet arrives at one of its interfaces. The decision-making table a router normally uses for applying this action is sometimes called the forwarding table and sometimes the routing table.

- **Error Control**

Error control also can be implemented in the network layer, the designers of the network layer in the Internet ignored this issue for the data being carried by the network layer.

➤ **Flow Control**

Flow control regulates the amount of data a source can send without overwhelming the receiver. If the upper layer at the source computer produces data faster than the upper layer at the destination computer can consume it, the receiver will be overwhelmed with data. To control the flow of data, the receiver needs to send some feedback to the sender to inform the latter that it is overwhelmed with data.

➤ **Congestion Control**

Another issue in a network-layer protocol is congestion control. Congestion in the network layer is a situation in which too many datagrams are present in an area of the Internet. Congestion may occur if the number of datagrams sent by source computers is beyond the capacity of the network or routers. In this situation, some routers may drop some of the datagrams.

• **Quality of Service**

As the Internet has allowed new applications such as multimedia communication (in particular real-time communication of audio and video), the quality of service (QoS) of the communication has become more and more important.

• **Security**

The network layer was designed with no security provision. Today, however, security is a big concern. To provide security for a connectionless network layer, we need to have another virtual

level that changes the connectionless service to a connection-oriented service. This virtual-layer, called **IPSec**.

IP – Internet Protocol

- It is the primary protocol in the Internet Layer of the Internet Protocol Suite .
- Messages are exchanged as data grams, also known as data packets or just packets.
- The main purpose and task of IP is the delivery of data grams from the source host to the destination host based on their addresses.

There are two versions of IP today

- IP version 4 (IPv4)
- IP version 6 (IPv6)

IPv4 ADDRESSES

An IPv4 address or simply IP address is a 32 bit address that uniquely and universally defines the connection of a device (for example: a computer or a router) to the internet.

ADDRESS SPACE

Address space is the total number of addresses used by the protocol. Of this number, IPv4 reserves some addresses for special purposes such as **private networks** (~18 million addresses) or **multicast addresses** (~270 million addresses).

The address space of IPv4 is 2^{32} or 4,294,967,296.

An IP address in **IPv4 is 32-bits in size**, which limits the address space to 4294967296 (2^{32}) IP addresses.

NOTATIONS

IPv4 addresses are usually represented in **dot-decimal notation**, consisting of four decimal numbers, each ranging **from 0 to 255**, separated by dots.

e.g., 172.16.254.1.

Each part represents a group of **8 bits (octet)** of the address.

In some cases of technical writing, IPv4 addresses may be presented in various **hexadecimal, octal, or binary** representations.

Binary notations

- In binary notation the IPv4 address is displayed as a 32 bits. Each octet is referred to as a byte.

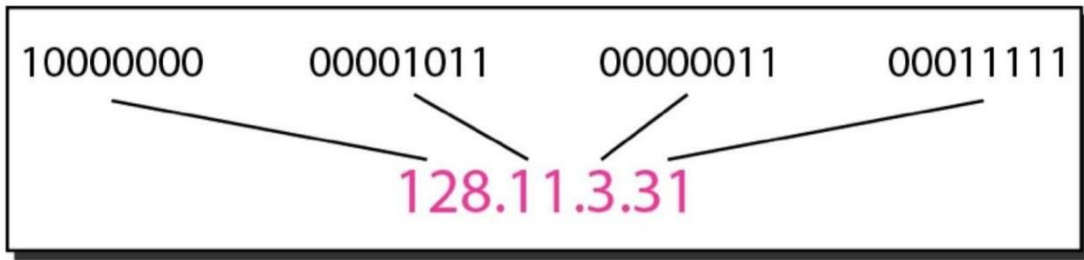
Dotted decimal notations

- To make the IPv4 address more compact and easier to read, we use decimal numbers with a decimal point separating bytes.

Hexa decimal notations

- IPv4 address in hexadecimal notation. Each hexadecimal digit is equivalent to four bits. This means 32-bit address has 8 hexadecimal digits.

Dotted-decimal notation and binary notation for an IPv4 address

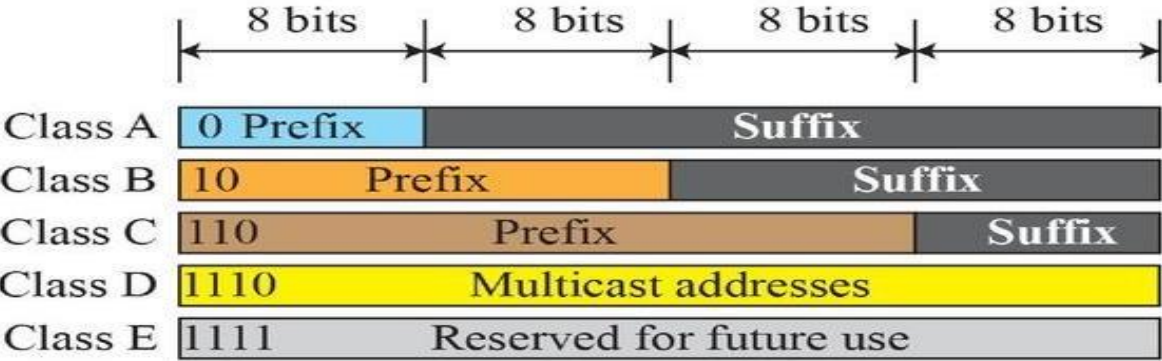
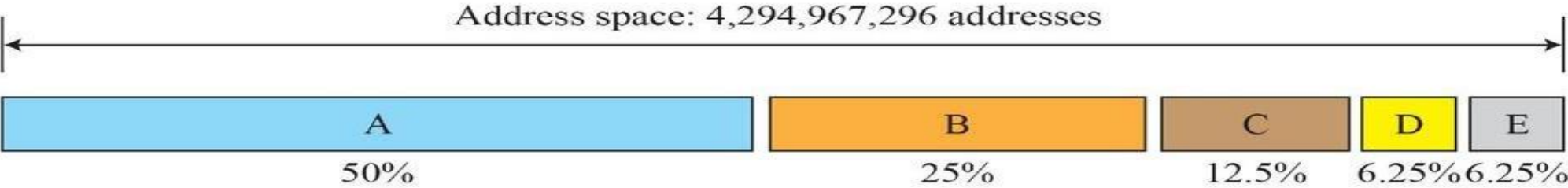


Hexadecimal notation : 800B031F

In 1981, the Internet addressing specification was revised with the introduction of **classful** network architecture. Broadly, the IPv4 Addressing system is divided into five classes of IP Addresses. All the five classes are identified by the first octet of IP Address. Internet Corporation for Assigned Names and Numbers (**ICANN**) is responsible for assigning IP addresses.

Classful Addressing

In classful addressing, the addressspace is divided into five classes: A,B,C,D and E.



Class	Prefixes	First byte
A	$n = 8$ bits	0 to 127
B	$n = 16$ bits	128 to 191
C	$n = 24$ bits	192 to 223
D	Not applicable	224 to 239
E	Not applicable	240 to 255

In class A, the network length is 8 bits ,but since the first bit, which is 0, defines the class, we can have only seven bits as the network identifier.This means there are only $2^7=128$ networks in the world that can have a class A address.

In class B, the network length is 16 bits ,but since the first two bits, which are $(10)_2$,defines the class, we can have only 14 bits as the network identifier.This means there are only $2^{14}=16,384$ networks in the world that can have a class B address.

In class C, the network length is 24 bits ,but since three bits define the class,we can have only 21 bits as the network identifier.This means there are $2^{21}= 2,097,152$ networks in the world that can have a class C address.

Class D is not divided into prefix and suffix.It is used for multicast address. All address that start with 1111 in binary belongs to **class E**. As in **class D**, **class E** is not divided into prefix and suffix and is used as reserve.

We can find the class of an address from the address. The classes and the specification is given below

Finding the classes in binary and dotted-decimal notation

	Octet 1	Octet 2	Octet 3	Octet 4		Byte 1	Byte 2	Byte 3	Byte 4
Class A	0.....				Class A	0–127			
Class B	10.....				Class B	128–191			
Class C	110.....				Class C	192–223			
Class D	1110....				Class D	224–299			
Class E	1111....				Class E	240–255			
	Binary notation					Dotted-decimal notation			

Find the class of each address.

- a.** 00000001 00001011 00001011 11101111
- b.** 11000001 10000011 00011011 11111111
- c.** 14.23.120.8
- d.** 252.5.15.111

Solution

- a.** The first bit is 0. This is a class A address.
- b.** The first 2 bits are 1; the third bit is 0. This is a class C address.
- c.** The first byte is 14; the class is A.
- d.** The first byte is 252; the class is E.

Number of blocks and block size in classful IPv4 addressing

<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

Classless Addressing / Classless Inter Domain Routing (CIDR)

As we have already learned about Classful Addressing, we are going to learn about **Classless Inter-Domain Routing**. which is also **known as Classless addressing**.

In the Classful addressing the no of Hosts within a network always remains the same depending upon the class of the Network.

Class A network contains 224 Hosts, Class B network contains 216 Hosts, Class C network contains 28 Hosts

Now, let's suppose an Organization requires 2^{14} hosts, then it must have to purchase a Class B network.

In this case, 49152 Hosts will be wasted. This is the major drawback of Classful Addressing. In order to reduce the wastage of IP addresses a new concept of **Classless Inter-Domain Routing** is introduced.

Whenever any user asks for IP addresses, IANA (Internet Assigned Numbers Authority, is an administrative function of the Internet that keeps track of IP addresses, domain names, and protocol parameter identifiers that are used by Internet standards.) is going to assign that IP addresses to the User.

Representation:

It is also a 32-bit address, which includes a special number which represents the number of bits that are present in the Block Id:

a . b . c . d / n

Where, n is number of bits that are present in Block Id / Network Id. Example:

20.10.50.100/20

CIDR (classless interdomain routing) notation is a compact representation of an IP address and its associated network mask. The notation was invented by **Phil Karn** in the 1980s. CIDR notation specifies an IP address, a **slash ('/') character**, and a decimal number.

The decimal number is the count of consecutive leading 1-bits (from left to right) in the

network mask.

The number of addresses inside a network or subnet may be calculated

as $2^{\text{address length} - \text{prefix length}}$, where *address length* is 32 for IPv4.

For example, in IPv4, the prefix length /29 gives: $2^{32-29} = 2^3 = 8$ addresses.

IPv6 ADDRESSES

The address depletion of IPv4 and other shortcomings of this (IPv4) protocol prompted a new version of IP protocol, which is called Internet Protocol Version 6(IPv6) Or IP new generation(IPng) was a proposal to augment the address space of IPv4 and at the same time redesign the format of the IP packet and revise some auxiliary protocols such as ICMP .(It is interesting to know that IPv5 was a proposal, based on the OSI the top model, that never materialized.

The following shows the main changes in Ipv6 protocol:

- 1.Larger address space: An IPv6 address is 128 bit long .
- 2.Better header format:IPv6 uses a new header format in which options are seperated from base header and inserted,when needed,between the base header and the data.

New options: IPv6 has new options to allow for additional functionalities.

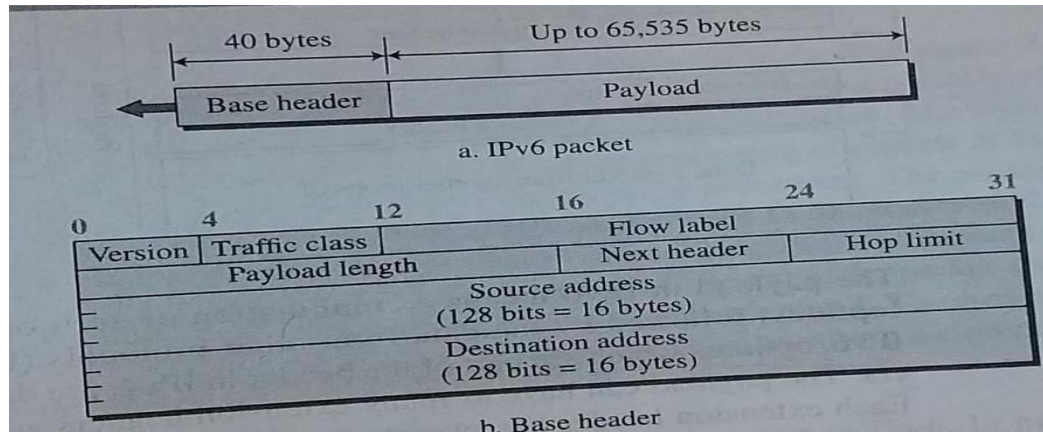
4.Allowance for extension: IPv6 is designed to allow the extension of the protocol if required by new technologies or applications.

5.Support for resource allocation:In IPv6,the type-of-service field is removed,but two new fields, traffic class and flow label, have been added.

6.Support for more security: Provide confidentiality and integrity of the packet.

Packet format

Each packet is composed of a base header followed by the payload.The base header occupies 40bytes, where as payload can be upto 65,535 bytes of information.



- *Version. The 4-bit version field defines the version number of the IP. For IPv6, the value is 6.
- *Traffic class. The 8-bit traffic class field is used to distinguish different payloads with different delivery requirements. It replaces the type-of-service field in IPv4.
- *Flow label. The flow label is a 20-bit field that is designed to provide special handling for a particular flow of data. We will discuss this field later.
- *Payload length. The 2-byte payload length field defines the length of the IP datagram excluding the header. Note that IPv4 defines two fields related to the length: header length and total length. In IPv6, the length of the base header is fixed (40 bytes); only the length of the payload needs to be defined.
- *Next header. The next header is an 8-bit field defining the type of first extension header (if present) or the type of the data that follows the base header in the datagram. This field is similar to the protocol field in IPv4, but we talk more about it when we discuss the payload.
- *Hop limit. The 8-bit hop limit field serves the same purpose as the TTL field in IPv4.
- *Source and destination address. The source address field is a 16-byte (128-bit) Internet address that identifies the original source of the datagram. The destination address field is a 16-byte (128-bit) Internet address that identifies the destination of the datagram.
- *Payload. Compared to IPv4, the payload field in IPv6 has a different format and meaning,

IPV6 ADDRESSING

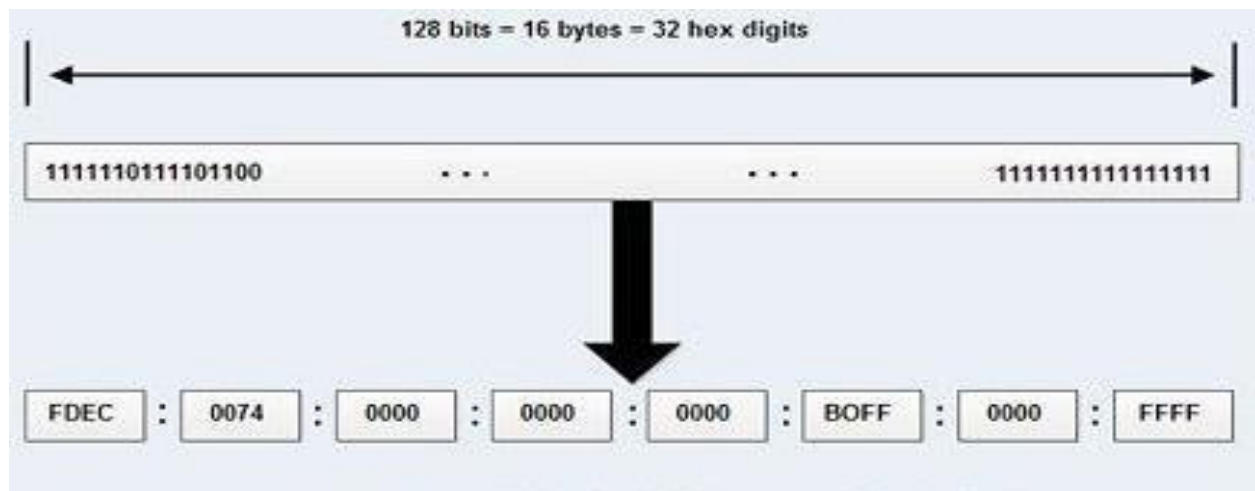
An IPv6 address is a 128 bit s or 16 bytes (octets) long, four times the address length in IPv4.

NOTATION

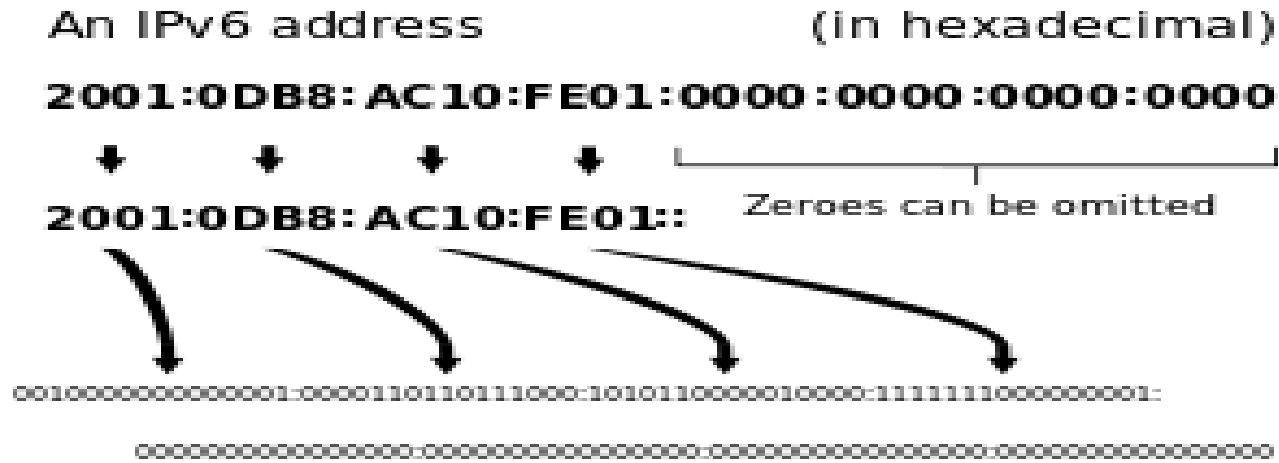
Binary Notation: which is used when the addresses are stored in a computer.

Colon Hexadecimal : In this notation ,128 bits is divided into eight sections, each 2 bytes in length. 2 byte hexadecimal notation requires four hexadecimal digits. So the address consists of 32 hexadecimal digits, with every four digits separated by a colon.

IPv6 address in binary and hexadecimal colon notation

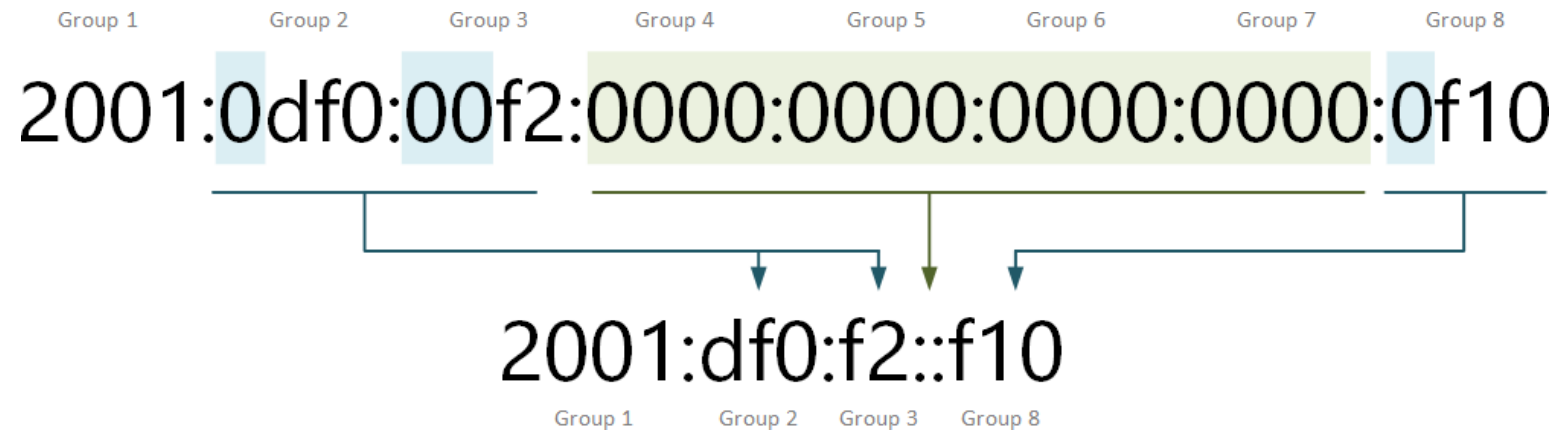


Abbreviation



- Although an IPv6 address, even in hexadecimal is very long, so we can abbreviate the address. The leading zeros of a section (four digits btwn two colons) can be omitted. Only the leading zeros can be dropped, not the trailing zeros.
- Using this form of abbreviation, 0074 can be written as 74, 000F as F and 0000 as 0. Note that 3210 cant be abbreviated.

- Further abbreviation is often called zero compression, can be applied to colon hexadecimal notation if there are consecutive sections consisting of zeros only. we can remove all the zeros altogether and replace them with a double semicolon.



```

34F6: 0000 : 0000 : 0000 : 0000 : 0001: 0095 : 9000
      ↓
34F6 :: 1 : 95 : 9000
  
```

Address Space

IPv6 has much larger address space; 2 power 128 addresses are available.

$2^{128} = 340,282,366,920,938,463,374,607,431,768,211,456$.

The three types of IPv6 addresses are: unicast, anycast, and multicast.

- **Unicast** addresses identify a single interface.
- **Anycast** addresses identify a set of interfaces in such a way that a packet sent to an anycast address is delivered to a member of the set.(delivered to most reachable member of the group)
- **Multicast** addresses identify a group of interfaces in such a way that a packet sent to a multicast address is delivered to all of the interfaces in the group.(each members will receive a copy)

Address space are divided for different purposes

A few left most bits called the type prefix, in each address define its category. The type prefix is variable in length, but it is unique one.

ROUTING ALGORITHMS

- The purpose of a routing algorithm is then simple: given a set of routers, with links connecting the routers, a routing algorithm finds a “good” path from source router to destination router.
- Typically a host is attached directly to one router; the default router for the host also called the first-hop router for the host and designation host as the destination router. **A graph** is used to formulate routing problems. Recall that a graph $G = (N, E)$ is a set N of nodes and a collection E of edges, where each edge is a pair of nodes from N .
- Nodes in the graph represent routers and edges connecting these nodes represent the physical links between these routers.
- An edge also has a value representing its cost.
- Since costs are assigned to the various edges in the graph abstraction, a natural goal of a routing algorithm is to identify the least costly paths between sources and destinations.
 - Least-cost path / shortest path is, the path with the smallest number of links between the source and the destination.
- ❖ **We can classify routing algorithms is according to whether they are global or decentralized**
- A **global routing algorithm** computes the least-cost path between a source and destination

using complete, global knowledge about the network. That is, the algorithm takes the connectivity between all nodes and all link costs as inputs.

- The algorithms with global state information are often referred to as **link-state (LS) algorithms**, since the algorithm must be aware of the cost of each link in the network.
- In a **decentralized routing algorithm**, the calculation of the least-cost path is carried out in an iterative, distributed manner. No node has complete information about the costs of all network links. Instead, each node begins with only the knowledge of the costs of its own directly attached links.
- The decentralized routing algorithm is called a **distance-vector (DV) algorithm**, because each node maintains a vector of estimates of the costs (distances) to all other nodes in the network.

❖ **Second Broad Way To Classify Routing Algorithm Is According To Whether They Are Static Or Dynamic.**

In **static routing algorithms**, route changes very slowly over time, often as a result of human intervention (for example, a human manually editing a router's forwarding table). In **Dynamic routing algorithms** change the routing paths as the network traffic loads or topology

change. A dynamic algorithm can be run either periodically or in direct response to topology or link cost changes. While dynamic algorithms are more responsive to network changes.

❖ The third way to classify routing algorithms is according to whether they are load-sensitive or load-insensitive.

In a **load-sensitive algorithm**, link costs vary dynamically to reflect the current level of congestion in the underlying link. High cost is associated with a link that is currently congested

The load-insensitive, does not explicitly reflect its current (or recent past) level of congestion

Link State Routing

Link state routing is a technique in which each router shares the knowledge of its neighborhood with every other router in the internetwork.

The three keys to understand the Link State Routing algorithm:

- **Knowledge about the neighborhood:** Instead of sending its routing table, a router sends the information about its neighborhood only.
- **Flooding:** Each router sends the information to every other router on the internetwork except its neighbors. This process is known as Flooding.
- **Information sharing:** A router sends the information to every other router only when the change occurs in the information.
- Recall that in a link-state algorithm, the network topology and all link costs are known, that is, available as input to the LS algorithm.
- This is accomplished by having each node broadcast link-state packets to *all* other nodes in the network, with each link-state packet containing the identities and costs of its attached links.
- The link-state routing algorithm is also known as *Dijkstra's algorithm*.

- **Dijkstra's algorithm computes the least-cost path from one node (the source, which we will refer to as u) to all other nodes in the network.**

Let us define the following notation:

$D(v)$: Cost of the least-cost path from the source node to destination v as its iteration of the algorithm.

$P(v)$: previous node (neighbor of v) along the current least cost path from the source to v .

N : subset of the nodes; v is in N if the least-cost path from the source to v is definitely known.

Distance Vector Routing Algorithm

- A distance-vector routing protocol requires that a router inform its neighbors of topology changes periodically.
- Compared to link-state protocols, which require a router to inform all the nodes in a network of topology changes, distance-vector routing protocols have less computational complexity and message overhead.
- The term distance vector refers to the fact that the protocol manipulates vectors (arrays) of distances to other nodes in the network.
- The vector distance algorithm was the original ARPANET routing algorithm and was also used in the internet under the name of RIP (Routing Information Protocol).

- Routers using distance-vector protocol do not have knowledge of the entire path to a destination. Instead they use two methods:
- 1. Direction in which router or exit interface a packet should be forwarded.
- 2. Distance from its destination

Distance-vector protocols are based on calculating the direction and distance to any link in a network.

- "Direction" usually means the next hop address and the exit interface. "Distance" is a measure of the cost to reach a certain node.
- The least cost route between any two nodes is the route with minimum distance.
- Each node maintains a vector (table) of minimum distance to every node.
- The cost of reaching a destination is calculated using various route metrics

- A **distance-vector routing (DVR)** protocol requires that a router inform its neighbors of topology changes periodically. Historically known as the old ARPANET routing algorithm (or known as Bellman-Ford algorithm).
- **Bellman Ford Basics** – Each router maintains a Distance Vector table containing the distance between itself and ALL possible destination nodes.
- Distances, based on a chosen metric, are computed using information from the neighbors' distance vectors.
- **Example** – Consider 3-routers X, Y and Z . Each router have their routing table. Every routing table will contain distance to the destination nodes.

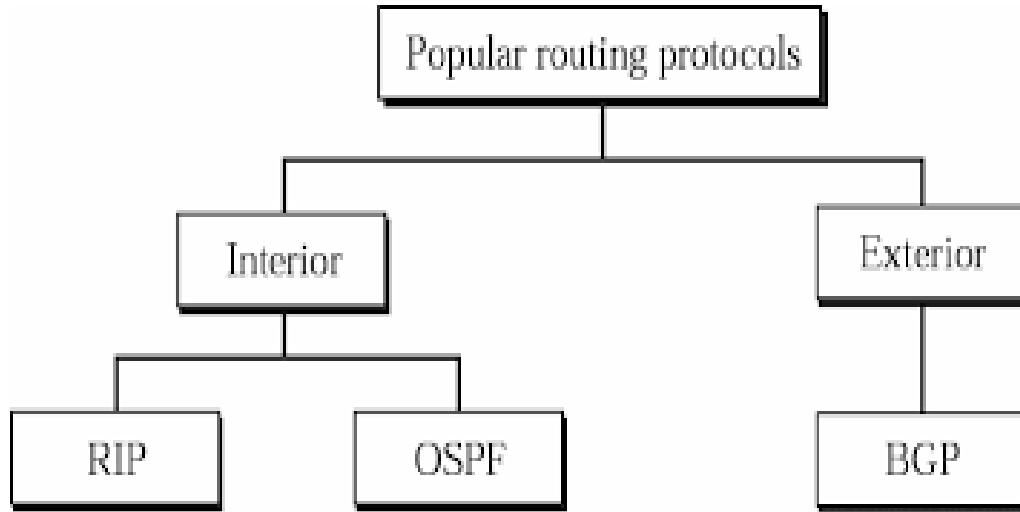
The Distance vector algorithm is iterative, asynchronous and distributed.

- **Iterative:** It is iterative in that its process continues until no more information is available to be exchanged between neighbors.
- **Asynchronous:** It does not require that all of its nodes operate in the lock step with each other.
- **Distributed:** It is distributed in that each node receives information from one or more

of its directly attached neighbours, performs calculation and then distributes the result back to its neighbours

- The Distance vector algorithm is a dynamic algorithm.
- It is mainly used in ARPANET, and RIP.
- Each router maintains a distance table known as **Vector**.

Routing on the internet



An internet can be so large that one routing protocol cannot handle the task of updating routing table of all routers .Thus, an internet is divided into autonomous systems (AS)

AS is a group of networks and routers under the authority of a single administration

Interior routing :Routing inside an autonomous system . Each AS can chose its own interior routing protocol

Examples: RIP and OSPF

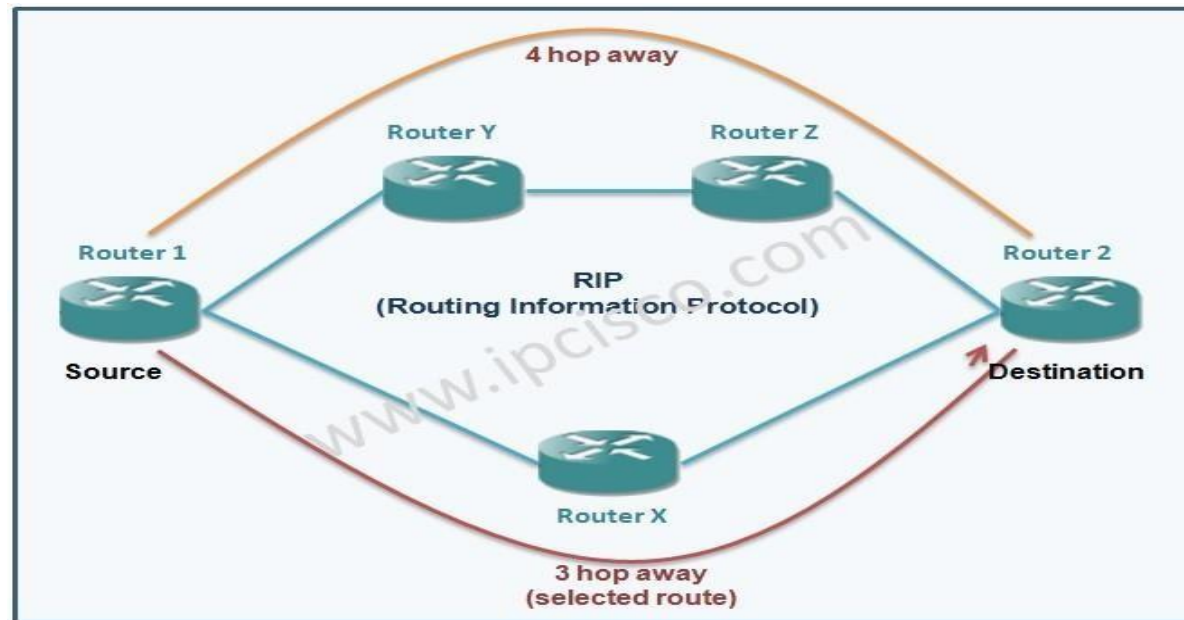
Exterior routing : Routing between autonomous systems . Only one exterior routing protocol is usually used for exterior routing .

Examples: BGP

For example, an organization can contain multiple routers having different locations, but the single autonomous number system will recognize them. Within the same autonomous system or same organization, we generally use IGP (Interior Gateway Protocol) protocols like RIP, IGRP, EIGRP, OSPF. Suppose we want to communicate between two autonomous systems. In that case, we use EGP (Exterior Gateway Protocols). The protocol that is running on the internet or used to communicate between two different autonomous number systems is known as BGP (Border Gateway Protocol). The BGP is the only protocol that is running on the internet backbone or used to exchange the routes between two different autonomous number systems. Internet service providers use the BGP protocol to control all the routing information.

Routing Information Protocol (RIP)

- RIP is a distance-vector routing protocol. Routers running the distance-vector protocol send all or a portion of their routing tables in routing-update messages to their neighbors. You can use RIP to configure the hosts as part of a RIP network.
- Which means that RIP is an older router protocol that uses hop count to determine the best route to send data packets to their destination.(the term hop refers to the total number of routers, from source to destination)



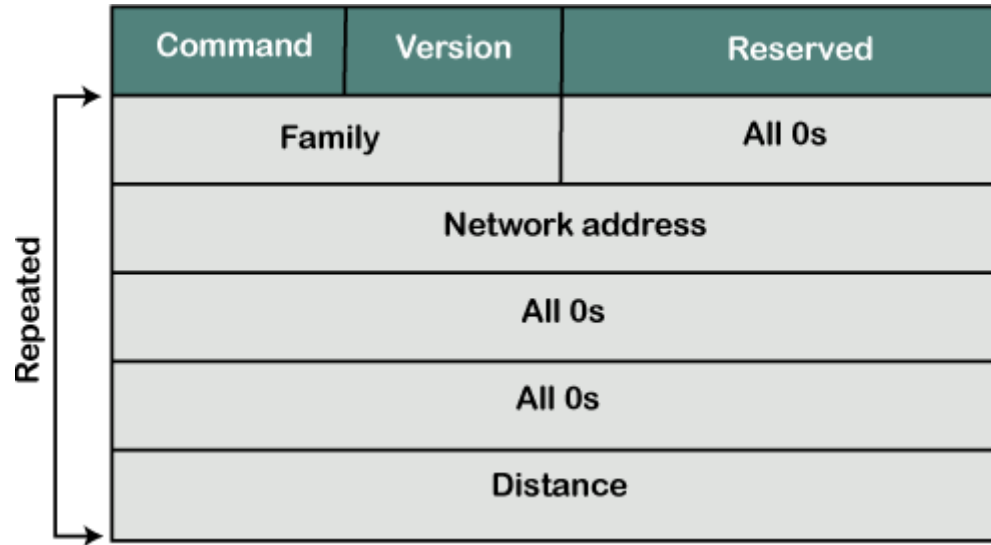
- RIP is an intradomain routing protocol used inside an autonomous system. The number of networks that the datagram can travel before it reaches its destination is defined as the metric. This metric is also known as the hop count.

In RIP ,routing updates are exchanged between neighbors approximately every 30seconds using RIP response message.

The response message send by a router or a host contain a list up to 25 destination subnets within the AS, as well as the sender's distance to each of those subnets. Response messages are also known as **RIP advertisements**.

RIP Message Format

The message format is used to share information among different routers. The RIP contains the following fields in a message:



- Command: It is an 8-bit field that is used for request or reply. The value of the request is 1, and the value of the reply is 2.
- Version: Here, version means that which version of the protocol we are using. Suppose we are using the protocol of version1, then we put the 1 in this field.
- Reserved: This is a reserved field, so it is filled with zeroes.
- Family: It is a 16-bit field. As we are using the TCP/IP family, so we put 2 value in this field.
- Network Address: It is defined as 14 bytes field. If we use the IPv4 version, then we use 4 bytes, and the other 10 bytes are all zeroes.

- Distance: The distance field specifies the hop count, i.e., the number of hops used to reach the destination.

Open Shortest Path First (OSPF)

- OSPF is a link-state routing protocol that was developed for IP networks and is based on the Shortest Path First (SPF) algorithm. OSPF is an Interior Gateway Protocol (IGP).
- OSPF divides an autonomous system into areas to handle routing efficiently and in a timely manner
- A collection of networks, hosts, and routers all contained within an autonomous system
- Thus, an autonomous system can be divided into many different areas
- All networks inside an area must be connected .Routers inside an area flood the area with routing information.
- Each area has a special router called area border routers
- Summarize the information about the area and sent it to other areas

- Among the area inside an autonomous system is a special area called backbone
- All of the areas inside an AS must be connected to the backbone

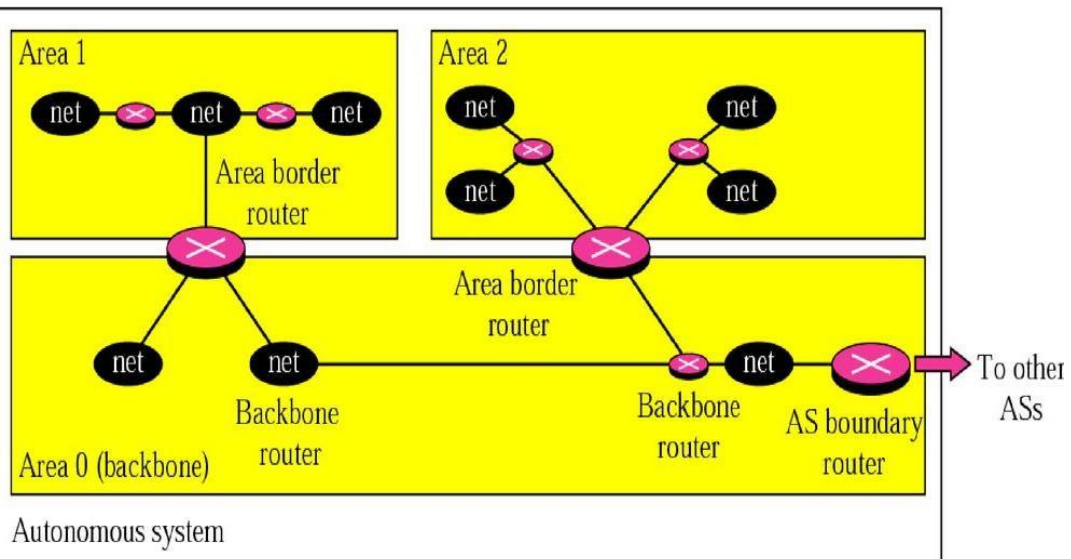
The routers inside the backbone are called the backbone routers

- A backbone router can also be an area border router
- If the connectivity between a backbone and an area is broken
- A virtual link must be created by the administration

Each area has an area identification

- The area identification of the backbone is zero

Areas in an Autonomous System



Metrics

- o OSPF allows the administrator to assign a cost, called the metric, to each route
- o Metric can be based on a type of service
 - a. Minimum delay
 - b. Maximum throughput
- o A router can have multiple routing tables
 - Each based on a different type of service

OSPF packet formats

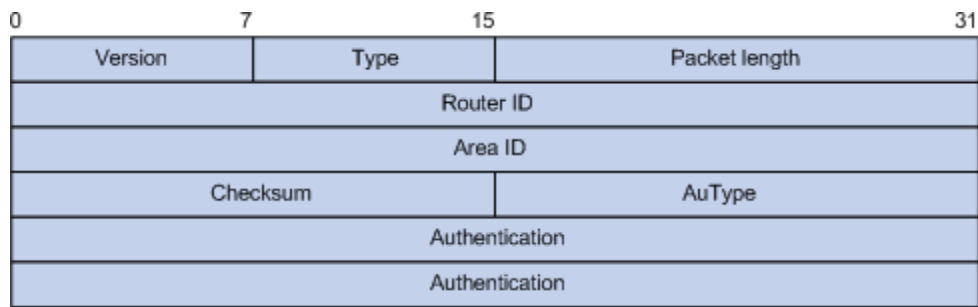
OSPF packets are directly encapsulated into IP packets. OSPF uses the IP protocol number 89. The format of an OSPF LSU packet (Link State Update (LSU) packets are OSPF packet Type 4.)

OSPF packet format

IP header	OSPF packet header	Number of LSAs	LSA header	LSA data
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OSPF packet header

OSPF packets are classified into five types that have the same packet header.



Major fields of the OSPF packet header are as follows:

- Version**—OSPF version number, which is 2 for OSPFv2.
- Type**—OSPF packet type from 1 to 5, corresponding to hello, DD , LSR, LSU, and LSAck, respectively.
- Packet length**—Total length of the OSPF packet in bytes, including the header.
- Router ID**—ID of the advertising router.
- Area ID**—ID of the area where the advertising router resides.
- Checksum**—Checksum of the message.
- AuType**—Authentication type, ranging from 0 to 2, corresponding to non-authentication, simple (plaintext) authentication, and MD5 authentication, respectively.
- Authentication**—Information determined by authentication type. It is not defined for authentication type 0. It is defined as password information for authentication type 1, and defined as Key ID, MD5 authentication data length, and sequence number for authentication type 2.

BGP(Border Gateway Protocol)

- o It is an inter-autonomous system routing protocol
- o Based on the path vector routing method

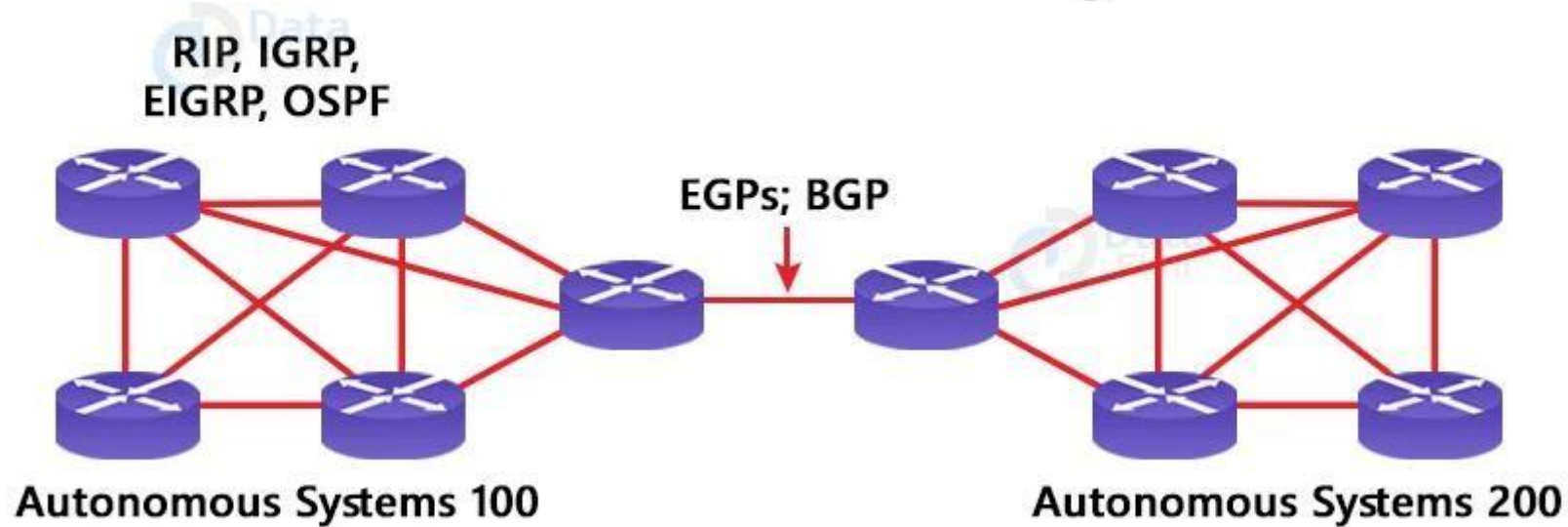
Border Gateway Protocol (BGP) refers to a gateway protocol that enables the internet to exchange routing information between autonomous systems (AS). As networks interact with each other, they need a way to communicate. This is accomplished through peering. BGP makes peering possible. Without it, networks would not be able to send and receive information with each other.

How does BGP work?

Each router maintains a routing table that controls how packets are directed. The BGP process on the router generates routing table information, which is based on the following factors:

- Incoming information from other routers.
- Information in the BGP routing information base (RIB), which is a data table stored on a server on the BGP router.

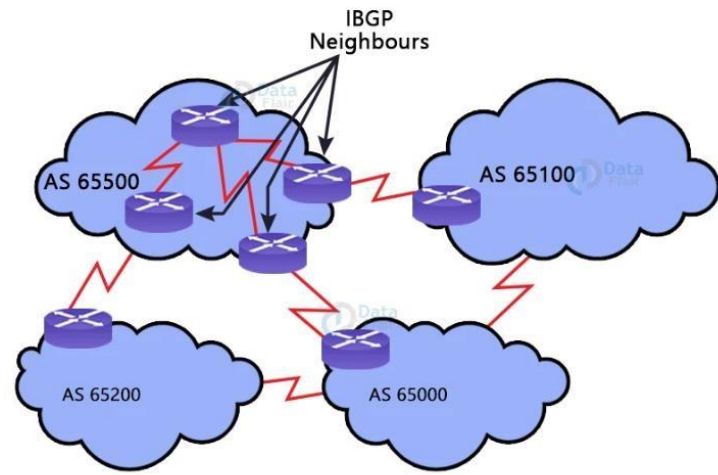
BGP Autonomous Systems



There are 2 types of BGP neighbour relationship:

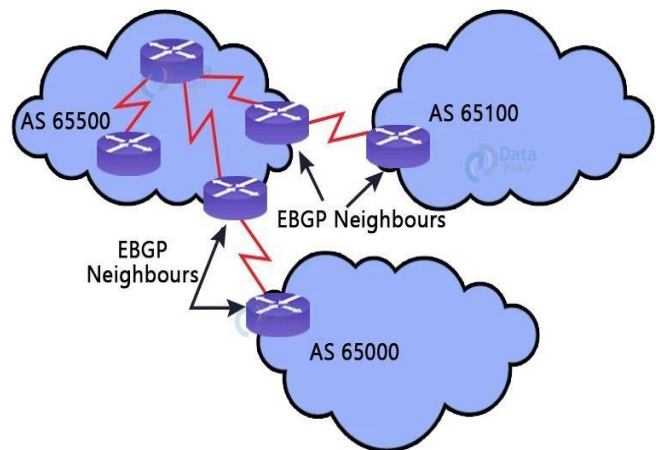
1. Internal BGP (IBGP):

The routers are referred to as an IBGP when they are all neighbours and belong to the same autonomous number system.



2. External BGP (EBGP):

The routers are referred to as an EBGP if they are all neighbours of each other and belong to different autonomous number systems.



Types of packets

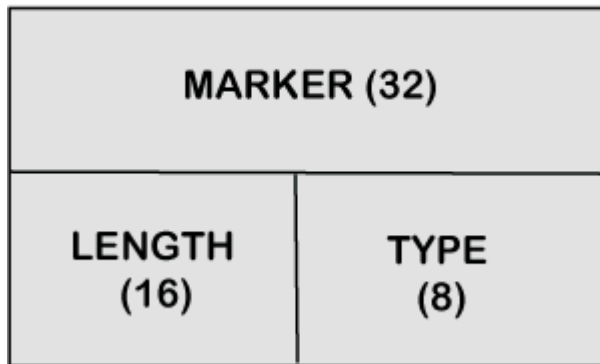
There are four different types of packets exist in BGP:

- **Open:** When the router wants to create a neighborhood relation with another router, it sends the Open packet.
- **Update:** The update packet can be used in either of the two cases:
 1. It can be used to withdraw the destination, which has been advertised previously.
 2. It can also be used to announce the route to the new destination.
- **Keep Alive:** The keep alive packet is exchanged regularly to tell other routers whether they are alive or not. For example, there are two routers, i.e., R1 and R2. The R1 sends the keep alive packet to R2 while R2 sends the keep alive packet to R1 so that R1 can get to know that R2 is alive, and R2 can get to know that R1 is alive.
- **Notification:** The notification packet is sent when the router detects the error condition or close the connection.

BGP Packet Format

Now we will see the format in which the packet travels. The following are the fields in a BGP packet format:

BGP Packet Format



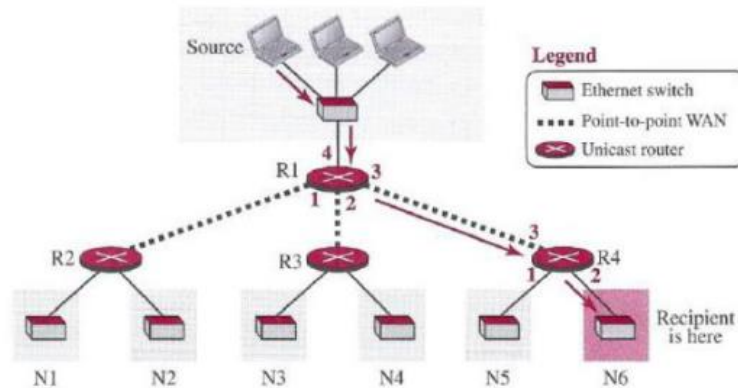
1. **Marker:** It is a 32-bit field which is used for the authentication purpose.
2. **Length:** It is a 16-bit field that defines the total length of the message, including the header.
3. **Type:** It is an 8-bit field that defines the type of the packet.

Multicast routing

In multicasting, there is one source and a group of destinations. The relationship is one to many. In this type of communication, the source address is a unicast address, but the destination address is a group address, a group of one or more destination networks in which there is at least one member of the group that is interested in receiving the multicast datagram. Multicasting starts with a single packet from the source that is duplicated by the routers. The destination address in each packet is the same for all duplicates. Note that only a single copy of the packet travels between any two routers.

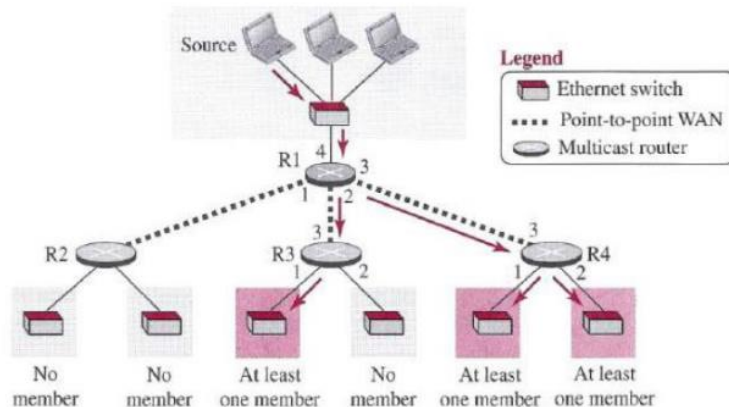
Unicasting

- There is one source and one destination network.
- The relationship between the source and the destination network is one to one
- Each router in the path tries to forward the packet to one and only one of its interfaces.



Multicasting

- There is one source and a group of destinations
- The relationship is one to many
- The source address is a unicast address, but the destination address is a group address, in which there is at least one member of the group that is interested in receiving the multicast datagram.



Congestion Control

congestion occurs in cases of traffic overloading when a link or network node is handling data in excess of its capacity.

- Congestion control is a method used for monitoring the process of regulating the total amount of data entering the network so as to keep traffic levels at an acceptable value.
- Congestion Window (cwnd) is a TCP state variable that limits the amount of data the TCP can send into the network before receiving an ACK. The Receiver Window (rwnd) is a variable that advertises the amount of data that the destination side can receive.

• Congestion Detection

TCP sender uses the occurrence of two events as signs of congestion in the network: time-out and receiving three duplicate ACKs.

If a TCP sender does not receive an ACK for a segment or a group of

segments before the time-out occurs, it assumes that the corresponding segment or segments are lost and the loss is due to congestion.

Another event is the receiving of three duplicate ACKs . Sending three duplicate ACKs is the sign of a missing segment, which can be due to congestion in the network.

Congestion Policies

TCP's general policy for handling congestion is based on three algorithms: slow start, congestion avoidance, and fast recovery.

Slow Start: Exponential Increase

The slow-start algorithm is based on the idea that the size of the congestion window (cwnd) starts with one maximum segment size (MSS), but it increases one MSS each time one acknowledgment arrives

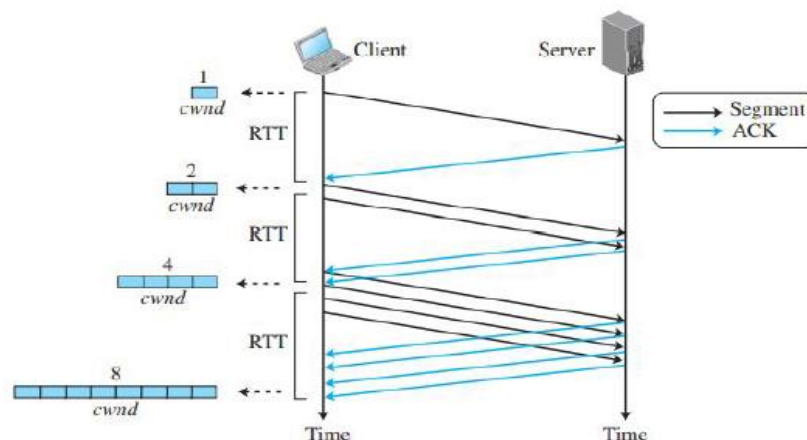
The sender starts with $cwnd = 1$. This means that the sender can send only one segment. After the first ACK arrives, size of the congestion window is increased by 1 because the arrival of the acknowledgment is a good sign

that there is no congestion in the network. The size of the window is now 2. After sending two segments and receiving two individual acknowledgments for them, the size of the congestion window now becomes 4, and so on.

In the slow-start algorithm, the size of the congestion window increases exponentially until it reaches a threshold.

Start	→	$cwnd = 1 \rightarrow 2^0$
After 1 RTT	→	$cwnd = cwnd + 1 = 1 + 1 = 2 \rightarrow 2^1$
After 2 RTT	→	$cwnd = cwnd + 2 = 2 + 2 = 4 \rightarrow 2^2$
After 3 RTT	→	$cwnd = cwnd + 4 = 4 + 4 = 8 \rightarrow 2^3$

Slow start, exponential increase



Congestion Avoidance: Additive Increase

TCP defines another algorithm called congestion avoidance, which increases the cwnd additively instead of exponentially.

- In this algorithm, each time the whole “window” of segments is acknowledged, the size of the congestion window is increased by one.
- In the congestion-avoidance algorithm, the size of the congestion window increases additively until congestion is detected.

Fast Recovery

- The fast-recovery algorithm is optional in TCP. The old version of TCP did not use it, but the new versions try to use it. It starts when three duplicate ACKs arrive that is interpreted as light congestion in the network. Like congestion avoidance, this algorithm is also an additive increase, but it increases the size of the congestion window when a duplicate ACK arrives.

QUALITY-OF-SERVICE (QOS)

- Quality of service (QoS) is an internetworking issue that refers to a set of techniques and mechanisms that guarantees the performance of the network to deliver predictable service to an application program.

Data-Flow Characteristics

- **Reliability.** Reliability is a characteristic that a flow needs in order to deliver the packets safe and sound to the destination. Lack of reliability means losing a packet or acknowledgment, which entails retransmission.
- **Delay.** Source-to-destination delay is another flow characteristic.
- **Jitter.** Jitter is the variation in delay for packets belonging to the same flow. For example, if four packets depart at times 0, 1, 2, 3 and arrive at 20, 21, 22, 23, all have the same delay, 20 units of time. On the other hand, if the above four packets arrive at 21, 23, 24, and 28, they will have different delays. For applications such as audio and video, the first case is completely acceptable; the second case is not.
- **Bandwidth.** Different applications need different bandwidths. In video conferencing we need to send millions of bits per second while the total number of bits in an e-mail may not reach even a million

Flow Classes

Based on the flow characteristics, classify flows into groups, with each group having the required level of each characteristic.

Flow Control to Improve QoS

1. Scheduling

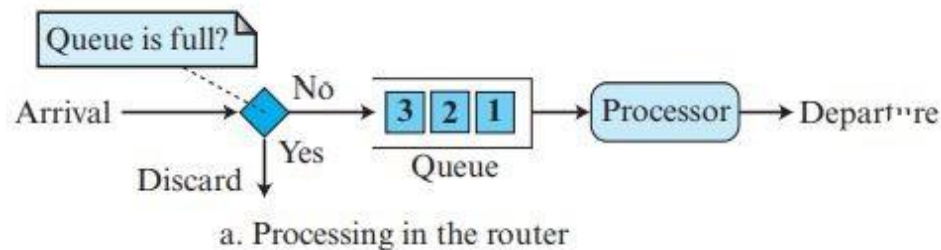
Several scheduling techniques are designed to improve the quality of service. They are

- FIFO queuing
- priority queuing
- weighted fair queuing

- FIFO Queuing

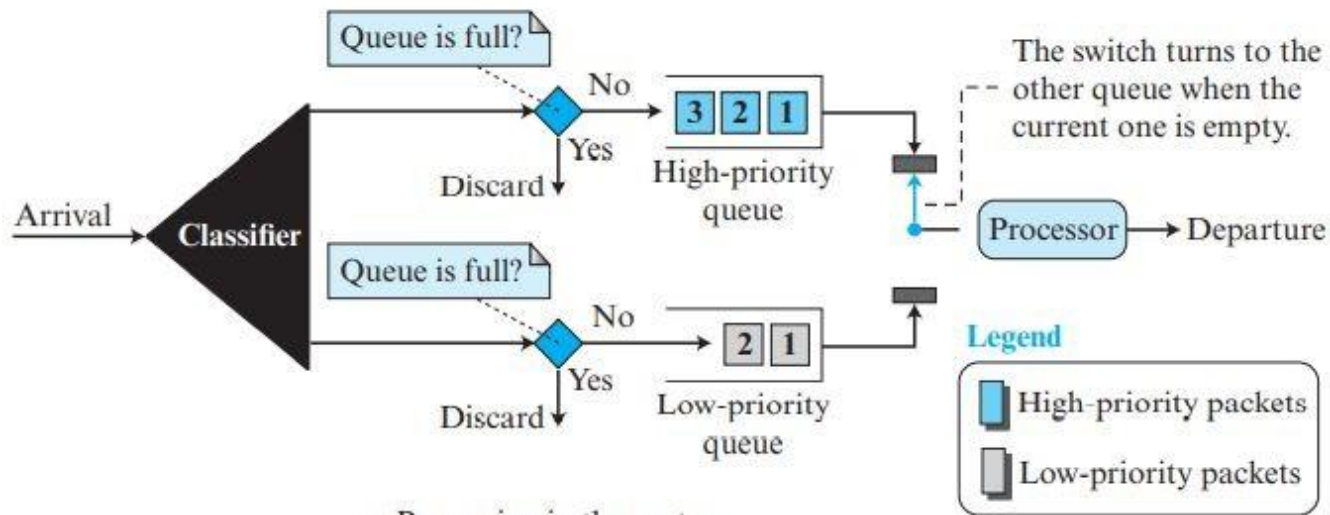
In first-in, first-out (FIFO) queuing, packets wait in a buffer (queue) until the node is ready to process them.

FIFO queuing is the default scheduling in the Internet. The only thing that is guaranteed in this type of queuing is that the packets depart in the order they arrive. This type of queuing is what we will see in the Internet with no service differentiation between packets from different sources.



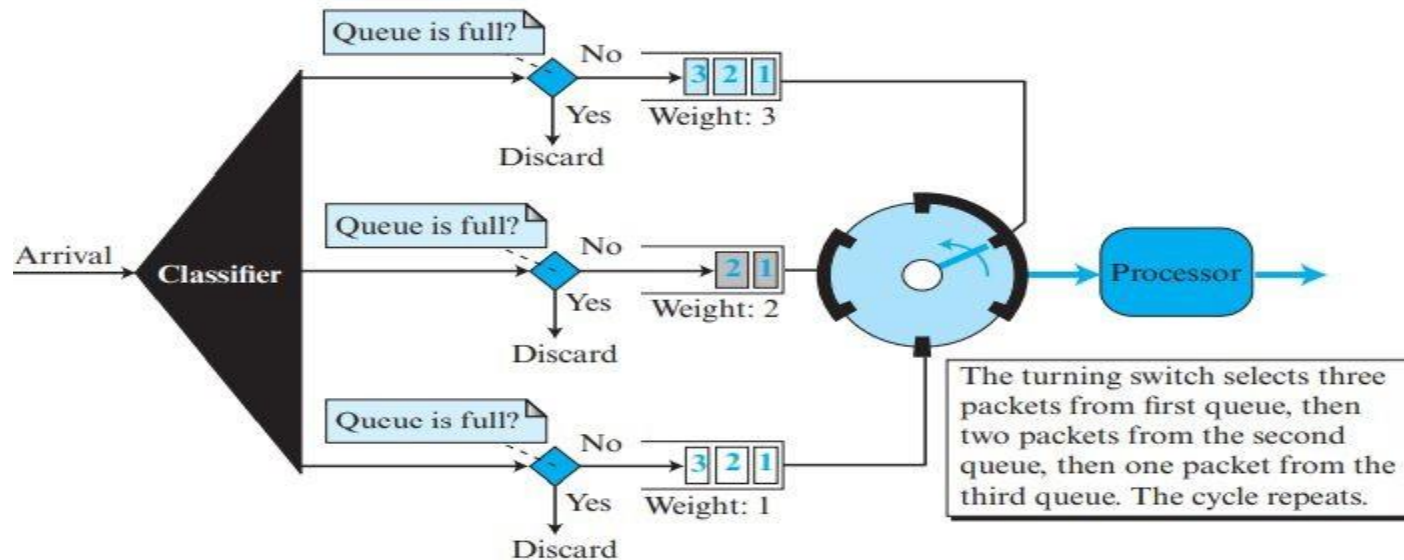
- Priority Queuing

In priority queuing, packets are first assigned to a priority class. Each priority class has its own queue. The packets in the highest-priority queue are processed first. Packets in the lowest-priority queue are processed last.



- Weighted Fair Queuing

A better scheduling method is weighted fair queuing. In this technique, the packets are still assigned to different classes and admitted to different queues. The queues however, are weighted based on the priority of the queues; higher priority means a higher weight. The system processes packets in each queue in a round-robin fashion with the number of packets selected from each queue based on the corresponding weight.

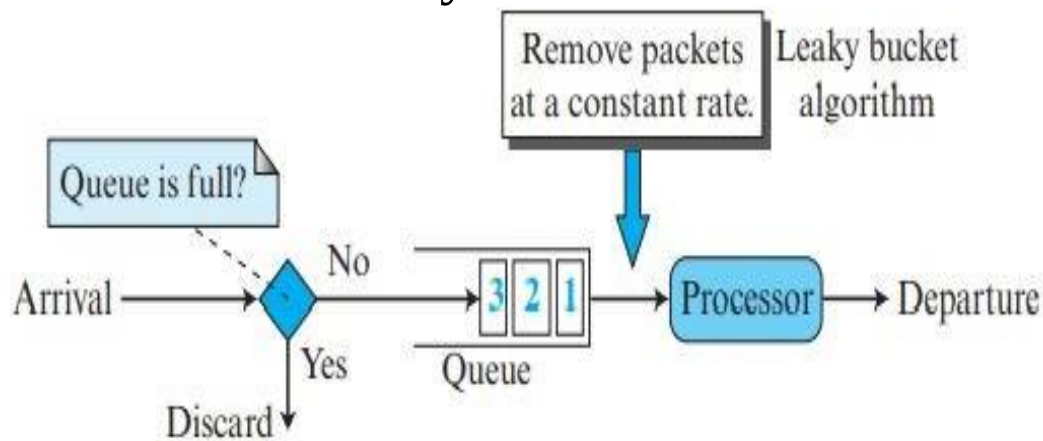


2. Traffic Shaping or Policing

To control the amount and the rate of traffic is called traffic shaping or traffic policing.

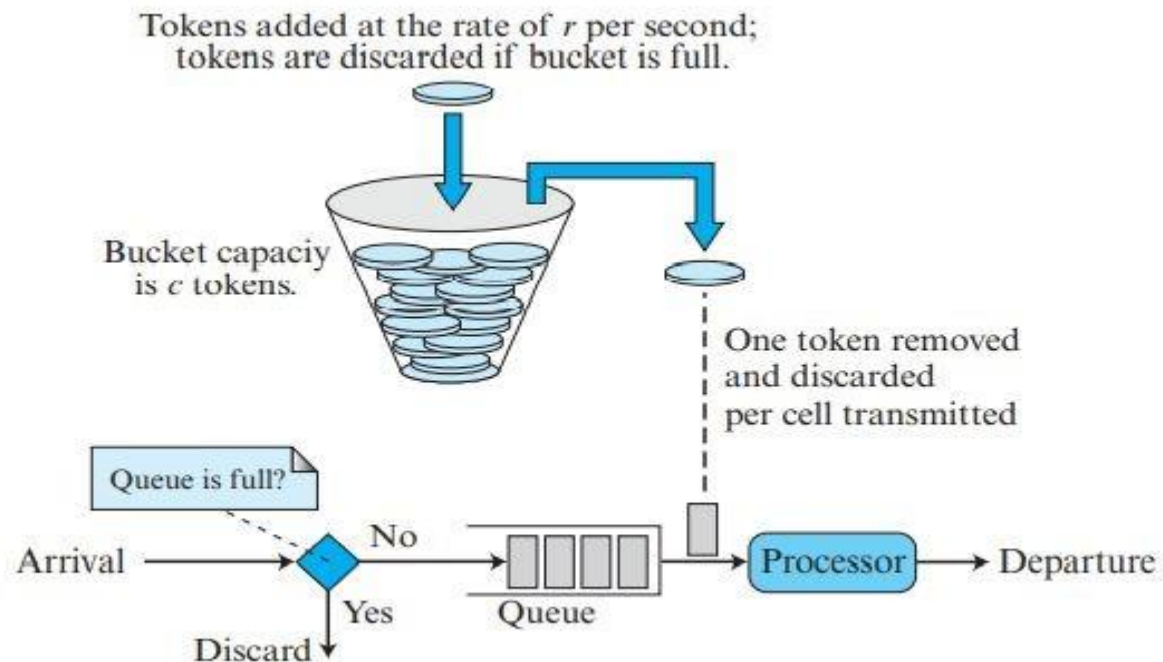
- Leaky Bucket

If a bucket has a small hole at the bottom, the water leaks from the bucket at a constant rate as long as there is water in the bucket. The rate at which the water leaks does not depend on the rate at which the water is input unless the bucket is empty. If the bucket is full, the water overflows. The input rate can vary, but the output rate remains constant. Similarly, in networking, a technique called leaky bucket can smooth out bursty traffic.



- Token bucket

Token bucket algorithm allows idle hosts to accumulate credit for the future in the form of tokens.



3.Resource Reservation

A flow of data needs resources such as a buffer, bandwidth, CPU time, and so on. The quality of service is improved if these resources are reserved beforehand.

4.Admission Control

Admission control refers to the mechanism used by a router or a switch to accept or reject a flow based on predefined parameters called flow specifications. Before a router accepts a flow for processing, it checks the flow specifications to see if its capacity can handle the new flow.
